



Bluetooth-Controlled Home Automation System

Team / Noha Ahmed Ibrahim Ahmed
Rahma Ashraf Elsayed Farghli

Track / Embedded Systems AVR

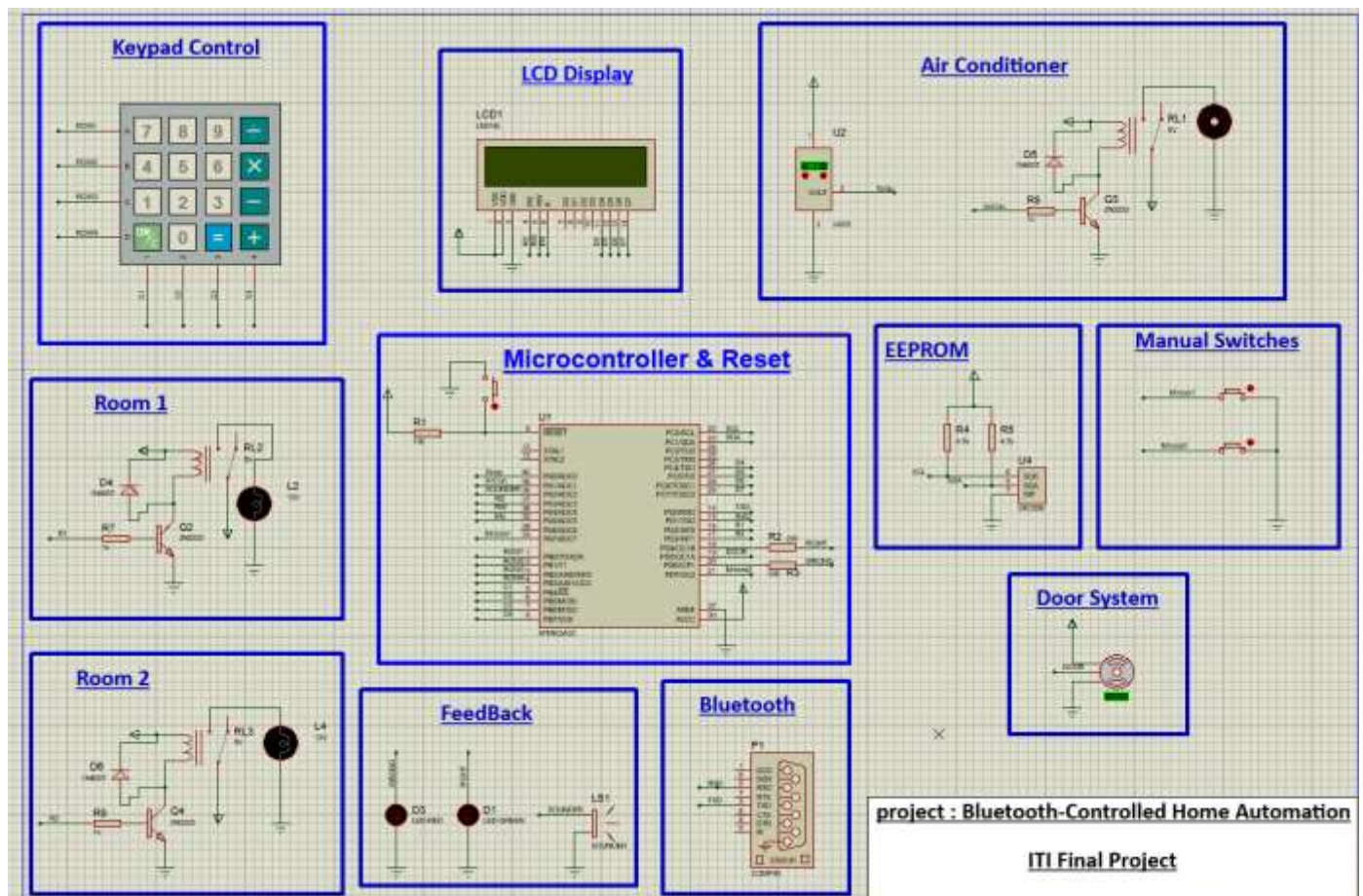
Class / 1

Instructors / Abdelrahman Okasha
Ahmad Mamdouh

Project Abstract:

This project presents a modular Smart Home Automation System utilizing the ATmega32 microcontroller to integrate security with environmental comfort. The system features a password-secured servo motor door lock, independent dual-room lighting, and automated LM35 temperature-controlled cooling. It utilizes an external I2C EEPROM for non-volatile storage and supports concurrent operation via local user inputs and wireless Bluetooth commands.

Schematic Diagram on Proteus Professional:



❏ **Software Architecture & Layered Design:**

1. Application Layer (APP): Contains the core logic

- a) Main.c
- b) HOME_Config.h
- c) HOME_Private.h
- d) HOME_Interface.h
- e) Home_Program.c

2. Hardware Abstraction Layer (HAL):

- a) LCD
- b) Keypad
- c) Servom
- d) Motor
- e) EEPROM
- f) Buzzer
- g) TempSensor
- h) LED

3. Microcontroller Abstraction Layer (MCAL):

- a)DIO
- b) ADC
- c)UART
- d)TWI
- e)Timer
- f)SPI
- g) EXTI

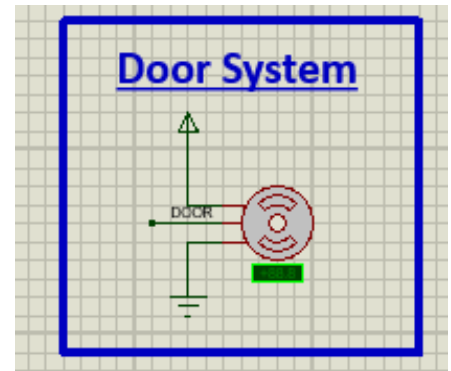
Hardware Components List:

1. **ATmega32 Microcontroller:** The main brain that controls the whole system and executes the code.
2. **LM016L (16x2 LCD):** Displays the system menus, temperature, and system status.
3. **Matrix Keypad:** Used by the user to enter the password and navigate menus.
4. **Servo Motor:** Simulates the smart door lock by rotating to open or close.
5. **LM35 Sensor:** Measures the ambient temperature for the automatic cooling system.
6. **24C02B (External EEPROM):** Stores the system password securely in non-volatile memory.
7. **COMPIM Module:** Simulates the serial/Bluetooth interface for wireless remote commands.
8. **SOUNDER (Buzzer):** Provides acoustic alerts for button presses and security alarms.
9. **Lamps (L2, L4):** Simulate the high-power lighting bulbs for Room 1 and Room 2.
10. **DC Motor:** Simulates the ventilation or cooling fan system.
11. **5V Relays:** Act as switches to safely control high-voltage lamps and fans using 5V signals.
12. **2N2222 Transistors:** Used as electronic switches to drive the relay coils.
13. **1N4007 Diodes:** Protect the circuit from voltage spikes generated by the relays.
14. **Push Buttons:** Used for direct manual control to toggle room lights.
15. **LEDs (Red & Green):** Provide visual feedback for success (Green) or system error (Red).
16. **Resistors:** Used for current limiting and pull-ups for communication lines.

□ System Features & Interfacing:

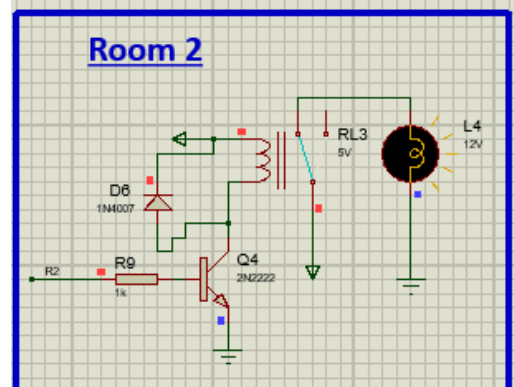
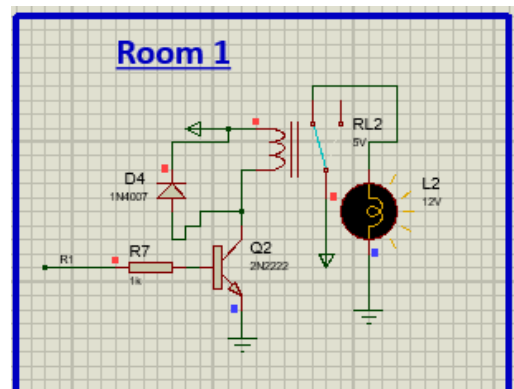
1. Password Security & Servo Door Lock

- **Proteus Block:** Keypad Control + Door System + Feedback
- **Interfacing Pins:** Keypad (PB0 to PB7), Servo Motor (PD5), Green LED (PD4), Red LED (PD6), Buzzer (PA2).
- **Func:** Enter a 4-digit password on the keypad to open the servo door. 3 wrong attempts trigger system lock.



2. Dual-Room Lighting Control

- **Proteus Block:** Room 1 + Room 2
- **Interfacing Pins:** Room 1 Relay (PD2), Room 2 Relay (PD3).
- **Func:** Switch lights ON/OFF independently for two separate rooms using relays and transistors.



3. Live Temperature Display

- **Proteus Block:** Air Conditioner (LM35 Sensor) + LCD Display



- **Interfacing Pins:** LM35 Sensor (PA0) | LCD Control (PA3, PA4, PA5) & LCD Data (PC4, PC5, PC6, PC7).
 - **Func:** Reads room temperature continuously via LM35 sensor and displays it dynamically on the LCD screen.
-
-

4. Manual Fan Management

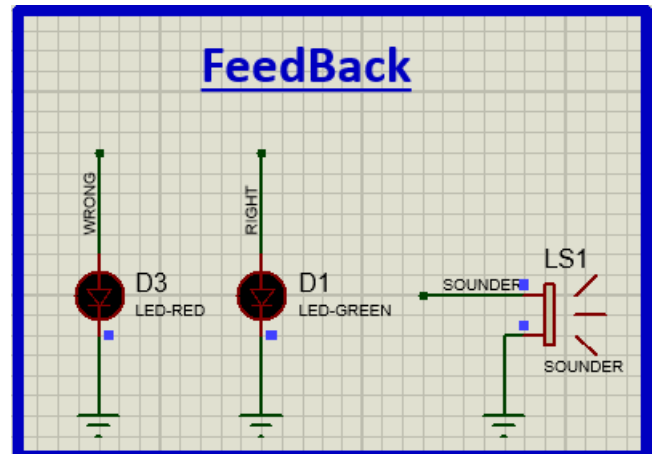
- **Proteus Block:** Air Conditioner (Relay RL1 & Fan) + LCD Display
 - **Interfacing Pins:** Fan Relay (PA1).
 - **Func:** Gives you full manual control to switch the cooling fan ON or OFF directly through the LCD menu or smartphone commands.
-
-

5. Smart Automatic AC Control

- **Proteus Block:** Air Conditioner (LM35 + Transistor Q3 + Relay RL1)
- **Interfacing Pins:** LM35 Sensor (PA0) & Fan Relay (PA1).
- **English:** Automatically switches the AC/Fan ON when the room temperature exceeds 30°C.

6. Visual & Audible Feedback

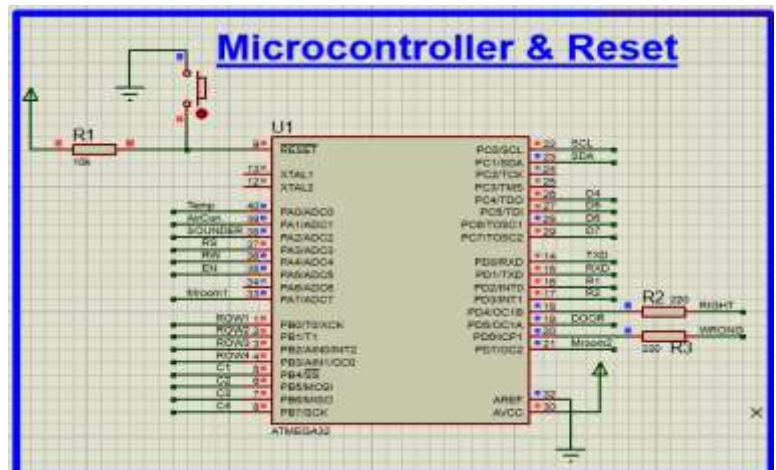
- **Proteus Block: Feedback (LED-GREEN, LED-RED, SOUNDER)**
- **Interfacing Pins: Green LED (PD4), Red LED (PD6), Buzzer (PA2).**



- **Func:** *Blinks Green LED for correct operations, and fires Red LED with Buzzer beeps for errors or wrong password entry.*

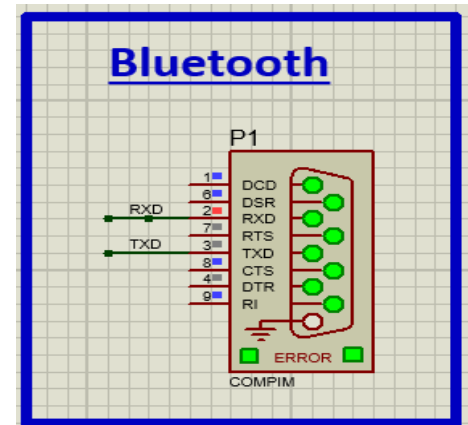
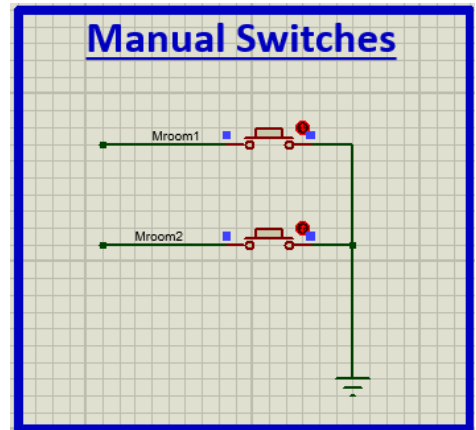
7. Full Hardware System Reset

- **Proteus Block:** Microcontroller & Reset (Reset Switch Button & R1)
- **Interfacing Pins:** Dedicated RESET Pin (Pin 9).
- **Func:** A dedicated push button to reboot the microcontroller and restart code execution instantly.



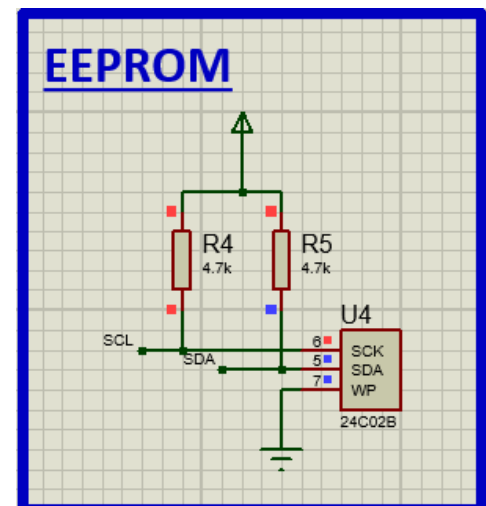
8. Hybrid Control (Local & Remote)

- **Proteus Block:** Bluetooth (COMPIM) + Manual Switches (Mroom1, Mroom2)
- **Interfacing Pins:** Bluetooth UART (PD0/RXD, PD1/TXD), Manual Switch 1 (PA7), Manual Switch 2 (PD7).
- **Func:** Allows controlling lights and fan locally using physical push buttons or remotely via Bluetooth mobile app.



9. EEPROM Data Logging & Mobile Report

- **Proteus Block:** EEPROM (24C02B) + Bluetooth (COMPIM)
- 3.
- **Interfacing Pins:** I2C TWI Pins (PC0/SCL, PC1/SDA) & Bluetooth UART (PD0/RXD, PD1/TXD).
- **English:** Saves event history inside external EEPROM memory during power loss, and sends a log report to your mobile via Bluetooth.





System Flowcharts :

